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Tank for transporting cryogenic fluids

Abstract

A tank (**12**) for transporting cryogenic fluids, comprising an outer container (**10**) and an inner container (**11**), having between them a compartment. The inner container (**11**) is adapted to contain a cryogenic fluid, which is brought outside the tank (**12**) through at least one cryogenic fluid withdrawal pipe (**13**), preferably made of stainless steel. The cryogenic fluid withdrawal pipe (**13**) is connected to the outer container (**10**) through a connection system (**22**) comprising a tubular bimetallic joint (**14**), which has an outer wall (**15**), adapted to be welded to the outer container (**10**) of the tank (**12**), and a central end or terminal (**16**), adapted to be welded to the cryogenic fluid withdrawal pipe (**13**).

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Background/Summary

(1) The present invention generally refers to a tank for transporting cryogenic fluids.
(2) The term cryogenic or cryogen fluid is often used to indicate substances that “generate cold” and those substances having a boiling temperature below −73° C. are considered as such.
(3) Substances falling into this family are, for example, nitrogen, helium, argon, carbon dioxide, oxygen, ammonia.
(4) Cryogenic fluids are usually transported inside tanks or containers at temperatures down to −196° C.; cryogenic tanks can be installed on refrigerated vehicles to feed their refrigeration system.
(5) Furthermore, to prevent the transported fluid from heating up, the tank or container must be isolated as much as possible from the external environment and for this reason the container is normally made up of two tanks, positioned one inside the other, between which it is created the vacuum.
(6) The fluid is contained in the inner tank and can come out through a pipe, which constitutes the fluid withdrawal pipe.
(7) Such pipe starts from the inner tank and passes through the outer tank.
(8) Other pipes start from the inner tank and pass through the outer tank with other functions, for example to fill the tank or to put the cryogenic fluid in communication with the system outside the tank.
(9) In the case of cryogenic tanks installed on vehicles, there is a need to reduce the weight of the tank as much as possible in order to consequently increase the capacity of the vehicle.
(10) Furthermore, especially in the case of tanks installed on refrigerated vehicles, there is a need to reduce the heat transmitted from the outer tank to the fluid withdrawal pipe.
(11) The aim of the present invention is to produce a tank for transporting cryogenic fluids, made of aluminum in order to reduce its weight as much as possible, which can be installed on a refrigerated vehicle to feed its refrigeration system.
(12) Another aim of the present invention is to produce a tank for transporting cryogenic fluids, which allows to reduce as much as possible the heat absorbed by the cryogenic fluid when it is withdrawn from the tank and therefore when it passes through the fluid withdrawal pipe.
(13) A further aim of the present invention is to produce a tank for transporting cryogenic fluids, which is particularly efficient, practical and safe.
(14) These and other aims are achieved by a tank for transporting fluids and, in particular,

cryogenic liquids, according to the attached claim **1**; other detailed technical characteristics of the tank object of the invention are specified in the subsequent dependent claims.

Description

(1) Further aims and advantages of the present invention will become clearer from the following description, relating to an exemplary and preferred, but not limiting, embodiment of the tank for transporting cryogenic fluids, according to the present invention, and from the attached figures, in which:
(2) FIG. **1** is a partial longitudinal sectional view of the tank for transporting cryogenic fluids, according to the present invention;
(3) FIG. **2** is a side view of the tank in FIG. **1**, according to the invention;
(4) FIG. **3** is an enlarged sectional view taken along the line A-A of the detail indicated with B in FIGS. **1** and **2**;
(5) FIG. **4** is an enlarged sectional view taken along the line A′-A′ of the detail indicated with C in FIGS. **1** and **2**;
(6) FIG. **5** is an enlarged and exploded view of the detail indicated with C in FIGS. **1** and **2**.
(7) With reference to the aforementioned figures, the tank **12** for transporting cryogenic fluids, according to the present invention, is advantageously made of aluminum and comprises an outer aluminum container **10** inside which an inner aluminum container **11** is arranged; between the outer container **10** and the inner container **11** there is a compartment inside which a vacuum can be created.
(8) The fluid is contained in the inner container **11** and can come out through at least one fluid withdrawal pipe **13**, preferably made of stainless steel.
(9) Other pipes or lines **13** are present to put the inside of the inner container **11** in communication with the outside of the tank **12**.
(10) A tank **12** of this type is advantageously used to be installed on a refrigerated vehicle to feed its refrigeration system and, in general, on refrigerated systems for industrial vehicles.
(11) According to the present invention, to reduce as much as possible the heat absorbed by the cryogenic fluid when it is withdrawn from the tank **12** and, therefore, when it passes through the fluid withdrawal pipe **13**, a particular system **22** is used (illustrated in detail in the attached FIG. **3**) to connect the withdrawal pipe **13** to the outer container **10**, reducing as much as possible the heat transmission.
(12) In particular, the aforementioned connection system **22** comprises a tubular bimetallic joint **14**, which has an aluminum end **15**, adapted to be welded to the outer container **10** of the tank **12**, and an end **16** made of stainless steel to be welded through the element of connection **17** to the fluid withdrawal pipe **13**.
(13) The connection system **22** includes a connection element **17** formed by concentric cones, which connects the bimetallic joint **14** to the fluid withdrawal pipe **13**, whose geometry is suitably designed to mechanically support the pipe **13**, reducing as much as possible the heat exchange with the external environment.
(14) Advantageously, the tank **12** also has a flanged connection **18** made with a first aluminum flange **19**, placed partially inside the compartment existing between the inner container **11** and the outer container **10**, and a second steel flange **20**, coupled to the first flange **19** and placed outside the outer container **10** (as shown in detail in the attached FIGS. **4** and **5**).
(15) The flanged connection **18** has multiple holes **21** for connecting several lines or pipes **13**; this allows with a single flanged connection **18** to pass through the outer container **10** with multiple lines or pipes **13** in order to put the inside of the tank **12** in communication with the outside.
(16) From the above description the characteristics of the tank for transporting cryogenic fluids,

which is the object of the present invention, are thus clear, as well as the advantages are clear.
(17) In particular, the essential advantages are the following: a tank for transporting cryogenic fluids made of aluminum; being able to install the tank on a refrigerated vehicle to feed the refrigeration system; being able to reduce as much as possible the heat absorbed by the cryogenic fluid when it is withdrawn from the tank; a flanged connection for the passage of cryogenic fluids with multiple holes for the passage of multiple lines or pipes towards the outside.
(18) Finally, it is clear that numerous other variations can be made to the tank in question, without thereby departing from the novelty principles inherent in the inventive idea, just as it is clear that, in the practical implementation of the invention, the materials, the shapes and the dimensions of the illustrated details may be any according to requirements and the same may be replaced with other equivalent ones.

Claims

1. A tank for transporting cryogenic fluids, comprising an external container inside which an internal container is arranged, so that a compartment is provided between said external container and said internal container, wherein said internal container is configured to contain a cryogenic fluid which is brought outside the tank through at least one cryogenic fluid withdrawal pipe for withdrawing said cryogenic fluid, wherein said at least one cryogenic fluid withdrawal pipe is connected to said external container by means of a connection system, characterized in that said connection system comprises a tubular bimetallic joint which has an aluminum end configured to be welded to said external container of the tank, and a stainless steel end, configured to be welded through a connection element to at least one cryogenic fluid withdrawal pipe.
2. The tank as per claim 1, wherein said connection system includes a tubular bimetallic joint with one end in aluminum and the other end in stainless steel.
3. The tank as per claim 1, wherein said connection system includes said connection element the latter being formed by concentric cones which connects said tubular bimetallic joint to said at least one cryogenic fluid withdrawal pipe, said connection element being configured to mechanically support said at least one withdrawal pipe minimize heat exchange with the environment outside said tank.
4. The tank as per claim 1, wherein said tank is installed on a refrigerated vehicle and is adapted to feed a refrigeration system of said vehicle.
5. The tank as per claim 1, wherein said tank has a flanged connection which includes a first flange placed partially inside said compartment, and a second flange coupled to said first flange and placed outside said outer container.
6. The tank as per claim 5, wherein said first flange is made of aluminum.
7. The tank as per claim 5, wherein said second flange is made of steel.
8. The tank as per claim 5, wherein said flanged connection has a plurality of holes for connecting a plurality of pipes or lines putting the cryogenic fluid in communication with the outside of the tank.
9. The tank as per claim 1, characterized by being made of aluminum.
10. Use of a tank as per claim 1 for refrigerated systems of industrial vehicles.